

Lab 1 – Version 1

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Version 1

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1. Introduction

Individuals with ADHD diagnoses or ADHD-like traits often struggle with consistent routine development. ADHD is medically classified into three subtypes: Inattentive, Hyperactive-Impulsive, and Combined (Chhabildas et al., 2001, 529). ADHD comes with productivity-altering traits including inattention, hyperactivity, impulsivity, memory problems, and a lack of focus (Staley et al., 2026, p. 890; Advanced Psychiatry Associates, 2026).

ADHD and its everyday challenges often persist beyond childhood. Routines may last anywhere from 3 days to “part of a month”, and are vulnerable to persistent Routine Adaptations (ADHD Philadelphia, 2026).

Routine Adaptations often lead to frustration, anxiety, avoidance, and reduced self-efficacy (Chen et al., 2026, pp. 2, 7). Self-reliance on Routine Adaptations is tedious: creating new schedules requires estimating uncertain timeframes. Routine Adaptations are subject to user error: individuals may forget to review their schedule, or unexpected Task Deviations may occur.

Gold Star is a proposed web-based habit-building and time management platform for users affected by ADHD. Gold Star is designed to help users organize schedules, break down tasks, and build sustainable routines. Gold Star’s intent is to make routine development easier to start, maintain, and complete daily. Gold Star’s prototype does not exist as of Version 1.

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2. Gold Star Product Description

Gold Star is a proposed web-based productivity and habit-building platform. Gold Star is primarily intended for individuals with ADHD who struggle with routine development and maintenance. Many people with ADHD struggle with organization, time management, and managing daily responsibilities.

Gold Star provides users a low-stakes environment for organizing and monitoring tasks, schedules, and habits without feeling overwhelmed. Gold Star is intended for easier routine development and maintenance. Gold Star offers support for users struggling to preserve long-term goal consistency.

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2.1. Key Product Features and Capabilities

Gold Star helps users organize tasks and schedules through an Adaptive Scheduling system which adjusts to missed tasks and changing priorities. Gold Star evolves with the user, reducing frustrations of falling behind, making it easier to stay on track.

Gold Star uses techniques like Habit Stacking, Activity Alternation, and the Pomodoro Technique which are recommended for people who struggle with ADHD-like symptoms.

Users can choose their productivity level and alter their schedule at any time, granting enough flexibility to feel successful. Users can group tasks by importance level, allowing sufficient personalization to suit users' scheduling needs.

Gold Star rewards users for completing tasks and milestones, giving them a sense of accomplishment. Gold Star does not punish users for missing goals, and it features a "Roll Over" system. The "Roll Over" system allows users to move missed tasks to another day.

Gold Star combines flexibility and structure by helping users overcome common ADHD challenges like task initiation, consistency, commitment, and completion. Overcoming such challenges leads to improving productivity and reducing daily stress.

Gold Star allows users to add and prioritize Rigid and Non-Rigid Tasks. Gold Star identifies and fills timeframes unoccupied by Rigid Tasks with recommended, prioritized Non-Rigid Tasks.

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2.2. Major Components (Hardware/Software)

Gold Star requires no specialized hardware. Users can access the application through standard commercial hardware with a modern web browser. The server side runs on a single hosting environment supporting Node.js and PostgreSQL, with deployments automatically managed through GitHub.

Figure 1 demonstrates Gold Star's three-tier structure: a Presentation Layer (web-based UI), an Application Layer (Node.js web server / API, authentication, and notification handler), and a Data Layer (PostgreSQL database storing encrypted user data, tasks, and routine data).

Gold Star's UI and UX are built with HTML, CSS, and JavaScript. Gold Star will feature a calendar view, routine dashboard, productivity level system, and statistics dashboard. Gold Star will feature a Node.js web server environment handling routing, logic, and authentication components. Gold Star will also provide a WebSocket-based notification handler and a PostgreSQL database. The database will store user account information, saved tasks, routine data, and encrypted sensitive user information.

Development is supported by off-the-shelf tools: Visual Studio Code, Git, GitHub, GitHub Actions for CI/CD, Jest for testing, and Javadoc for documentation.

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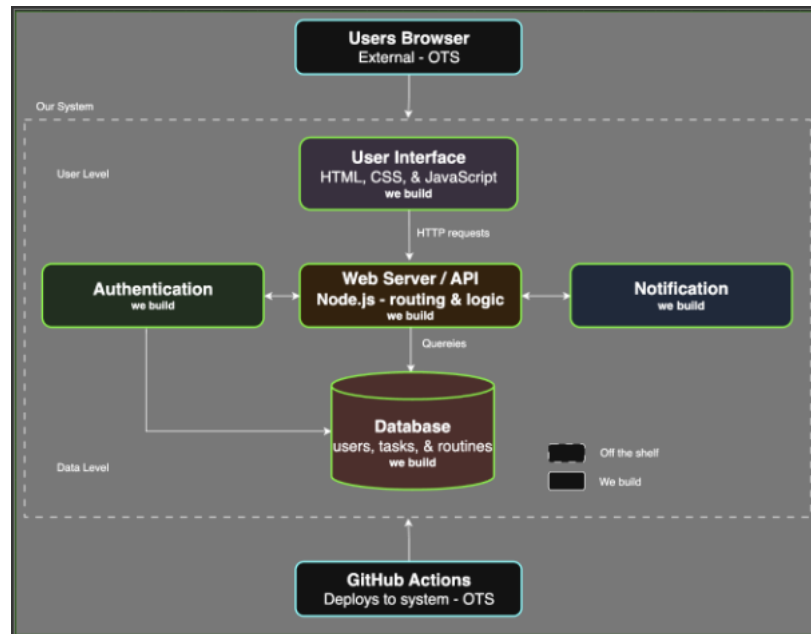


Figure 1: Gold Star's Major Functional Component Diagram

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3. Glossary

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. Example: a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Non-Rigid Task – Tasks lacking required starting times and deadlines. Non-Rigid Tasks require an estimated timeframe for completion. Non-Rigid Tasks are scheduled by Adaptive Scheduling.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. Example: a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Rigid Task – Non-negotiable tasks requiring a specific starting time or deadline for completion.

Rigid Tasks are scheduled by Rigid Scheduling.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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4. References

- ADHD Philadelphia. (2026, May 30). *Why Adults With ADHD Struggle to Stay Consistent With Routines*. ADHD Philadelphia. <https://www.adhdphiladelphia.com/blog/why-adults-with-adhd-struggle-to-stay-consistent-with-routines>
- Advanced Psychiatry Associates. (2026). *Strategies for Adults Living With ADHD*. Addicta. <https://advancedpsychiatryassociates.com/resources/blog/strategies-for-adults-living-with-adhd>
- CDC. (2024, May 29). *ADHD Across the Lifetime*. Attention-Deficit / Hyperactivity Disorder (ADHD). <https://www.cdc.gov/adhd/articles/adhd-across-the-lifetime.html>.
- Chen, J., Meng, Y., & Nie, K. (2026, March). “*Not Just Me and My To-Do List*”: Understanding Challenges of Task Management for Adults with ADHD and the Need for AI-Augmented Social Scaffolds. arXiv. <https://doi.org/10.48550/arXiv.2603.17258>
- Chhabildas, N. Pennington, B.F. & Willcutt, E.G. (2001, December). A Comparison of the Neuropsychological Profiles of the DSM-IV Subtypes of ADHD. *Journal of Abnormal Child Psychology*, 29, 529-540. <https://doi.org/10.1023/A:1012281226028>
- Kessler, R.C., Lane, M., Stang, P.E., & Van Brunt, D.L. (2008, April 21). The prevalence and workplace costs of adult attention deficit hyperactivity disorder in a large manufacturing firm. *Psychological Medicine*, 39(1), 137-147. <https://doi.org/10.1017/S0033291708003309>
- Kreider, C. M., Medina, S., & Slamka, M. R. (2019). Strategies for Coping with Time-Related and Productivity Challenges of Young People with Learning Disabilities and Attention-

Deficit/Hyperactivity Disorder. *Children (Basel, Switzerland)*, 6(2), 28.

<https://doi.org/10.3390/children6020028>

Ogrodnik, M., Karsan, S., Malamis, B., Kwan, M., Fenesi, B., & Heisz, J. J. (2023). Exploring Barriers and Facilitators to Physical Activity in Adults with ADHD: A Qualitative Investigation. *Journal of developmental and physical disabilities*, 1–21. Advance online publication. <https://doi.org/10.1007/s10882-023-09908-6>

Staley, B.S., Robinson, L.R., Claussen, A.H., Katz, S.M., Danielson, M.L., Summers, A.D., Farr, S.L., Blumberg, S.J., & Tinker, S.C. (2024). Attention-Deficit/Hyperactivity Disorder Diagnosis, Treatment, and Telehealth Use in Adults — National Center for Health Statistics Rapid Surveys System, United States, October–November 2023. *U.S. Centers for Disease Control and Prevention Morbidity and Mortality Weekly Report*, 70(40), 890–895. <https://www.cdc.gov/mmwr/volumes/73/wr/pdfs/mm7340a1-H.pdf>

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5. Placeholder Table Section

Table 1 is a replica of Gold Star’s Real-World Product vs. Prototype table. This section is provided to refer at least one table for Version 1.

Features & Functionality	Real-World Product	Prototype (Planned)
Calendar View	✓	✓
Adaptive Scheduling	✓	✓
Task Prioritization	✓	✓
Habit Stacking	✓	✓
Reward System	✓	✗
Customizable GUI	✓	✗

Table 1: Real-World Product vs. Prototype

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